

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: a0999e39dd9521f6

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_64_qam.iq
File SHA-256 Digest : a0999e39dd9521f6be0cb94dcce382ab202a9deccaf674b91894801f5f7e8e0f
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T19:48:18Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	5.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+378.190 kHz	Spectral Centroid Offset
Estimated Baud Rate	49.29 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	2.44 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	731.20 kHz	Cumulative PSD 99% Energy
Estimated SNR	3.46 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	6.73 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+299987.79 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0043	Wiener Entropy Index (0.0 – 1.0)

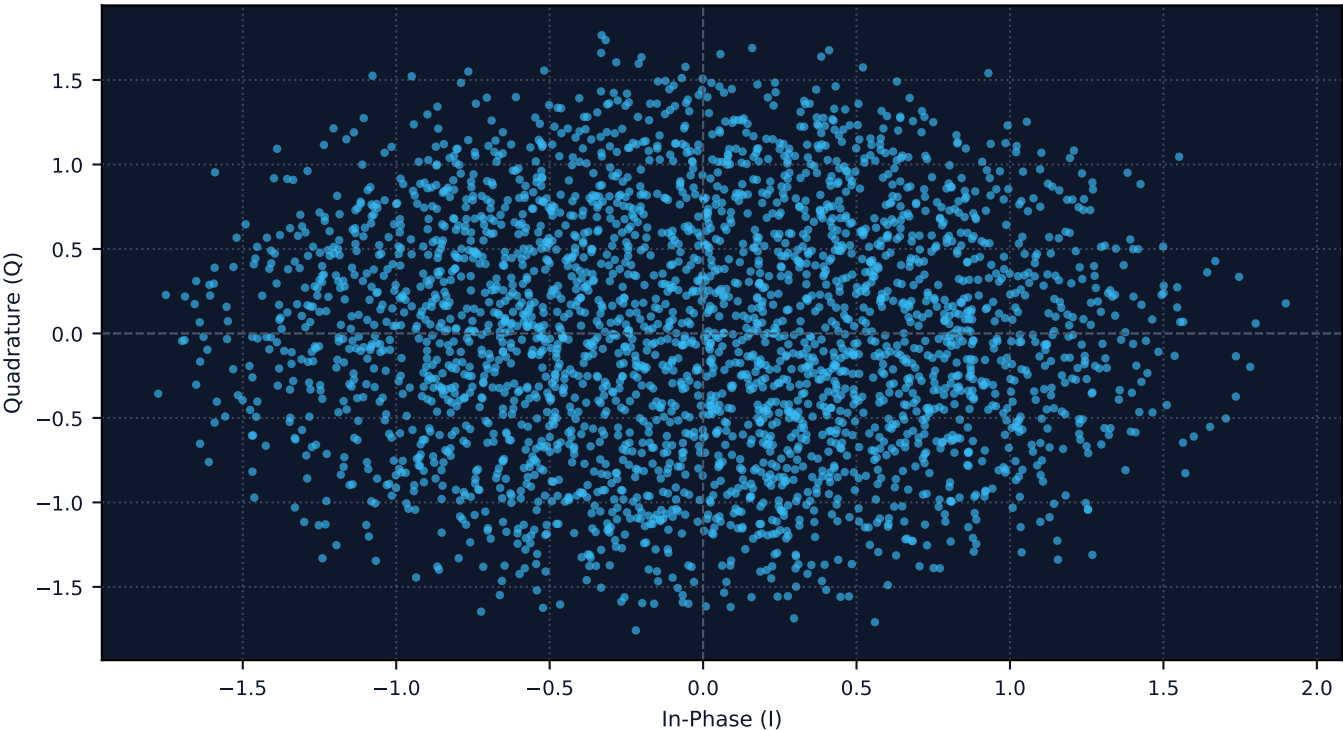
PREDICTED MODULATION : 64-QAM

CLASSIFIER CONFIDENCE : 72.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : WARNING: LOW CONFIDENCE CLASSIFICATION

Demodulation Scheme: 64-QAM | Recovered Bits: 393,216 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [64-QAM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	133	0x0000	0	20813 B	CRC MISMATCH	CRC_MISMATCH
#2	195	0x0000	1	20026 B	CRC MISMATCH	CRC_MISMATCH
#3	1027	0x0000	2	49016 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1102	0x0000	3	17751 B	CRC MISMATCH	CRC_MISMATCH
#5	1239	0x0000	4	23667 B	CRC MISMATCH	CRC_MISMATCH

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 0228C23867DAC7161C50D2628EB1FEE7 [FRAME 1 PAYLOAD]